

Program : Diploma in Automation and Robotics	
Course Code : 5333C	Course Title: IoT and Applications
Semester : 5	Credits : 4
Course Category : Program core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To study fundamental concepts of IoT
- To Learn different protocols used for IoT design.
- Understand the role of IoT in various domains of Industry.

Course Prerequisites:

- Basic knowledge in Data Communication, Computer Networks

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic concepts of IoT and IoT networking.	14	Understanding
CO2	Perceive ideas on the different connectivity technologies and Wireless Sensor Networks.	14	Understanding
CO3	Explain the basic concepts of cloud computing and fog computing	15	Understanding
CO4	Apply the basics of IoT for different applications like Smart Homes and Industrial IoT.	15	Understanding
	Series Test	2	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1							
CO2	2	2					
CO3	2	2					
CO4		2					

3-Strongly mapped,2-Moderately mapped,1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic concepts of IoT and IoT networking.		
M1.01	Define IoT	6	Understanding
M1.02	Enlist any three characteristics of IoT and explain.	8	Understanding
Contents: Introduction to Internet of Things Introduction Characteristics of IoT- Applications of IoT- IoT categories- IoT Enablers and Connectivity Layers - Baseline Technologies-Sensors-Characteristics of a Sensor Classification of Sensors-Actuators-Types of Actuators-IoT components and implementation. Service Oriented Architecture-IoT Interdependencies-Challenges for IoT. IoT Networking Connectivity Terminologies, Gateway Prefix Allotment-Impact of Mobility on Addressing Multihoming-Deviations from Regular Web-IoT Identifications and Data Protocols.			
CO2	Perceive ideas on the different connectivity technologies and Wireless Sensor Networks.		
M2.01	Discuss on Plasma HART and Wireless HART.	8	Understanding
M2.02	Explain the challenges in WSN	6	Understanding
	Series Test – I	1	
Contents: Connectivity Technologies Introduction-IEEE 802.15.4-ZigBee-6LoWPAN-RFID-HART and wireless HART-NFC Bluetooth-Z-wave-ISA 100.11A. Wireless Sensor Networks Introduction-Components of a sensor Node-Modes of Detection-Challenges in WSN-Sensor Web-Cooperation-Behaviour of Nodes in WSN-Information Theoretic Self-Management of WSN-Social Sensing in WSN-Applications of WSN-Wireless Multimedia Sensor Networks. Wireless Nano sensor Networks-Under Water Acoustic Sensor Networks-WSN Coverage. Optimal Geographical Density Control (OGDC) Algorithm-Stationary WSN-Mobile WSN.			

CO3	Explain the basic concepts of cloud computing and fog computing		
M3.01	Explain public cloud.	8	Understanding
M3.02	List out the advantages of Fog.	7	Understanding
Contents: Cloud Computing Introduction-Architecture-Characteristics-Deployment Models-Public Cloud-Private Cloud-Hybrid Cloud-Community Cloud-Multi Cloud-Distributed Cloud-Inter Cloud-Big Data Cloud-HPC Cloud-Service Models-Service Management-Cloud Security. Fog Computing Introduction-Why Fog Computing-Requirements of IoT-Architecture of Fog-Working of Fog-Advantages of Fog-Applications of Fog-Challenges in Fog.			
CO4	Apply the basics of IoT for different applications like Smart Homes and Industrial IoT.		
M4.01	Understand non-industrial applications of IoT	7	
M4.02	Understand industrial applications of IoT	8	
	Series Test – II	1	
Contents: IoT Applications Smart Homes Introduction-Origin of Smart Home-Examples of Smart Home Technologies-Smart Home Implementation-Home Area Networks-HAN Elements-HAN Standards-HAN Architectures -HAN Initiatives-Smart Home Benefits and Issues. Industrial IoT Introduction-IIoT Requirements-Design Considerations-Applications of IIoT-Manufacturing Industry-Health Care Service Industry-Transportation and Logistics-Mining-Firefighting-Smart Dust-Drones-Futuristic Farming-Aerospace-Energy Networks-Benefits of IIoT-Challenges of IIoT.			

Text / Reference

T/R	Book Title/Author
T1	Honbo Zhou, “The Internet of Things in the Cloud: A Middleware Perspective”, CRC Press, 2012
T2	Jeeva Jose, “Internet of Things”, Khanna Book Publishing Co.(P) Ltd, 1/e
R1	Hanes David ,Salgueiro Gonzalo , Grossetete Patrick , Barton Rob , Henry Jerome, ‘IoT Fundamentals: Networking Technologies, Protocols and Use Cases for the Internet of Things’, Pearson, 1/e.
R2	Rajkamal, “Internet of Things: Architecture and Design Principles”, McGraw Hill (India) Private Limited
R3	Daniel Minoli, — “Building the Internet of Things with IPv6 and MIPv6: The Evolving World of M2M Communications”, ISBN: 978-1-118-47347-4, Willy Publications